

INDU 211 · Term Paper Group Presentation Guide & 15-Minute Video Script

"The Future of Industrial Engineering in the Generative AI Era"

Concordia University · Department of Mechanical, Industrial and Aerospace Engineering

Course: INDU 211 / Section X · Due Date: Tuesday, December 1, 2026, 11:59 PM

Format: 15-Minute Recorded Video Presentation (Zoom / Teams / OBS Studio)

1. Executive Presentation Specifications & Timing Architecture

15-MINUTE VIDEO RECORDING BENCHMARKS	
Target Duration	Exactly 15:00 minutes (±30 seconds tolerance)
Team Participation	Every team member MUST present an equal share
Delivery Format	Screen-share with active speaker webcams on
Video Container	.MP4 or .AVI (H.264 1080p, 30fps recommended)
Audio Standard	Clear vocal clarity with minimal background hum

1.1 Team Speaker Distribution (5 Members · 3:00 min each)

SPEAKER ALLOCATION MATRIX		
Window	Speaker	Presentation Theme
00:00 – 03:00 m	Speaker 1 (Mohamed S.)	Title, Executive Motivation, Industrial Paradigms & Taxonomic Foundations
03:00 – 06:00 m	Speaker 2	Workplace Transformation & Automation Potential Across IE Sub-Disciplines
06:00 – 09:00 m	Speaker 3	Critical Failure Modes, Hallucinations, Physical Ungroundedness & Security Risks
09:00 – 12:00 m	Speaker 4	Industry 5.0 & The Human-in-the-Loop Systems Architect Role
12:00 – 15:00 m	Speaker 5	Graduate Competency Roadmap, Strategic Career Blueprint & Q&A Synthesis

2. Slide-by-Slide Verbatim Presentation Script

Slide 1: Title & Team Introductions (00:00 – 01:00)

- **Slide Visual:** Professional title card with Concordia MIAE emblem, full paper title, course designation, and team member headshots/IDs.
- **Speaker 1 (Mohamed S.):**

"Good afternoon, Professor and fellow engineers. Welcome to our presentation on 'The Future of Industrial Engineering in the Generative AI Era: Autonomous Operations, Human-in-the-Loop Synergies, and Next-Generation Competency Paradigms'. My name is Mohamed Sharafath, and I am joined today by my colleagues [Names]. Together, we represent the Apex Industrial Analytics Group. Over the next fifteen minutes, we will present a comprehensive, evidence-based investigation into how transformer foundation models, generative agents, and automation algorithms are reshaping modern manufacturing—and why the human industrial engineer is more essential today than ever before."

Slide 2: The Industrial Engineering Paradigm & GenAI Inflection (01:00 – 02:00)

- **Slide Visual:** Timeline graphic from Taylor's scientific management (Industry 1.0) to CIM (Industry 3.0), IoT (Industry 4.0), and the GenAI disruption.
- **Speaker 1 (Mohamed S.):**

"For over a century, Industrial Engineering has been the foundational science of efficiency, waste reduction, and systems optimization. From Taylor's time studies to linear programming and Six Sigma, our discipline has always integrated machinery, people, and information. However, the rise of Generative AI represents a qualitative leap. Unlike traditional automated systems that follow rigid, deterministic scripts, GenAI possesses the ability to synthesize natural language, generate functional code, formulate candidate production schedules, and propose facility layouts in seconds."

Slide 3: Taxonomic Clarification: AI vs. ML vs. GenAI (02:00 – 03:00)

- **Slide Visual:** Concentric Venn diagram and functional table contrasting Classical AI (rule-based/MILP), Predictive ML (Regression, SVM), and GenAI (LLMs, Diffusion).
- **Speaker 1 (Mohamed S.):**

"To evaluate this transformation rigorously, we must dispel the myth that AI is a single monolith. Classical Operations Research uses exact solvers like Gurobi to find global optima for linear constraints. Predictive Machine Learning forecasts remaining equipment life or customer demand based on historical patterns. But Generative AI models probability distributions to create entirely new engineering artifacts—from bill-of-materials databases to simulation models. I will now hand over to [Speaker 2] to examine how these capabilities impact specific IE sub-disciplines."

Slide 4: IE Automation Feasibility-Impact Matrix (03:00 – 04:30)

- **Slide Visual:** Clean, high-contrast matrix displaying the 6 IE sub-disciplines ranked by automation feasibility and impact (from Table 1 of report).
- **Speaker 2:**

"Thank you, Mohamed. As industrial engineers, our first analytical inquiry is: which core functions can be automated, and which require human oversight? Our research established a Feasibility-Impact Matrix across six major domains. Production scheduling and supply chain forecasting demonstrate the highest automation potential—exceeding seventy-five percent—because they rely heavily on pattern recognition and combinatorial heuristics. Conversely, ergonomics and physical facility layout design exhibit significantly lower automation feasibility—between thirty and fifty percent—because they are fundamentally constrained by human biomechanics and real-world spatial physics."

Slide 5: Production Scheduling & Line Balancing Automation (04:30 – 06:00)

- **Slide Visual:** Comparison diagram of a classic Gantt chart alongside an autonomous generative dispatching agent dynamically adjusting to machine breakdown.
- **Speaker 2:**

"Consider production scheduling. In a complex job-shop environment, solving NP-hard sequencing problems manually or via basic heuristics is time-consuming. GenAI paired with deep reinforcement learning can generate optimized dispatching schedules in seconds. Similarly, in assembly line balancing, algorithms can rapidly partition precedence graphs to equalize workstation cycle times. However, complete autonomy remains an illusion. When a critical machine blows a hydraulic seal, or when rush orders arrive without notice, generative algorithms lack the contextual business awareness to negotiate multi-objective trade-offs. Now, [Speaker 3] will address the critical failure modes of these autonomous systems."

Slide 6: Hallucinations & The Absence of Physical Grounding (06:00 – 07:30)

- **Slide Visual:** Schematic showing an LLM generating ungrounded CNC feeds causing tool breakage versus a verified physically-constrained model.
- **Speaker 3:**

"Thank you. While the generative capacity of foundation models is impressive, their deployment in safety-critical manufacturing introduces severe vulnerabilities. The most dangerous flaw is probabilistic hallucination. Large Language Models predict token probabilities; they possess zero innate understanding of Newton's laws of motion, thermodynamic heat dissipation, or structural material limits. In production trials, LLMs have confidently recommended CNC cutting feed rates that would instantly shatter carbide tooling and cause catastrophic spindle collisions. In physical engineering, a hallucination is not an inconvenience—it is a life-safety hazard."

Slide 7: Deterministic Violations, Black-Boxes & IP Security (07:30 – 09:00)

- **Slide Visual:** Graphic showing a circular dependency error in precedence constraints alongside cybersecurity data leakage icons (CAD blueprints leaking to public clouds).
- **Speaker 3:**

"Furthermore, generative models struggle with hard deterministic constraints. In line balancing, task precedence is absolute—task B cannot start before task A is completed. Yet generative transformers frequently produce circular precedence loops because they attend to semantic similarity rather than mathematical topology. Beyond logic errors, industrial enterprises face acute cybersecurity risks. Ingesting proprietary CAD models or proprietary chemical recipes into external cloud-based GenAI platforms creates intolerable intellectual property exposure. Under ISO 9001 and aerospace AS9100 standards, unexplainable 'black-box' algorithms cannot sign off on quality certifications. I now pass the floor to [Speaker 4] to discuss the human role in mitigating these risks."

Slide 8: Industry 5.0 & The Human-in-the-Loop Paradigm (09:00 – 10:30)

- **Slide Visual:** Conceptual illustration of Industry 5.0: The human engineer at the center, surrounded by autonomous digital twins and AI agents.
- **Speaker 4:**

"Thank you. The critical limitations just described illuminate why the future belongs to Industry 5.0. While Industry 4.0 prioritized blind machine-to-machine automation, Industry 5.0 re-centers the human being at the core of the industrial ecosystem. In this paradigm, the Industrial Engineer evolves from a manual calculator into a high-level Cyber-Physical Systems Architect. Through a Human-in-the-Loop, or HITL framework, the engineer uses GenAI as an analytical exoskeleton—automating eighty percent of nominal, repetitive tasks while reserving human cognition for edge cases, ethics, and physical validation."

Slide 9: Systemic Constraint Formulation & Socio-Technical Leadership (10:30 – 12:00)

- **Slide Visual:** Flow diagram demonstrating the engineer defining the boundary conditions (X , Y constraints) and the AI performing high-speed design space exploration.
- **Speaker 4:**

"AI can optimize within a box, but it cannot define the box. The human engineer's primary mandate is systemic constraint formulation: translating ambiguous business goals, environmental sustainability targets, and labor contracts into rigorous mathematical objective functions. Moreover, manufacturing plants are deeply socio-technical organizations. When automation is introduced, floor technicians often experience anxiety and resistance. Industrial engineers are uniquely qualified to design human-centered workplaces that prioritize psychological safety, physical ergonomics, and meaningful work. I will now hand over to [Speaker 5] to outline how graduates must prepare for this future."

Slide 10: The Next-Gen IE Graduate Skills Pyramid (12:00 – 13:30)

- **Slide Visual:** The 4-tier Graduate Skills Pyramid from Section 6 of the report (Foundations → Computation → Digital Twins → Leadership).
- **Speaker 5:**

"Thank you. To lead this transformation, future industrial engineering graduates must cultivate a robust, T-shaped skill set, represented by our Next-Gen Skills Pyramid. At the foundational base, classic IE principles remain non-negotiable: deterministic operations research, Queuing theory, and Statistical Quality Control. On top of this, graduates must build computational fluency: mastering Python for industrial analytics, utilizing Google OR-Tools and Gurobi APIs, and mastering prompt engineering and Retrieval-Augmented Generation to anchor LLMs to verified enterprise databases. Moving up the pyramid, digital twin mastery using Siemens Tecnomatix and NVIDIA Omniverse allows us to stress-test AI solutions in physics-accurate virtual sandboxes before committing physical capital."

Slide 11: Techno-Economic Acumen & Change Management (13:30 – 14:15)

- **Slide Visual:** Balance scale graphic comparing AI operational investment (capex/opex) against proven productivity yield and risk mitigation.
- **Speaker 5:**

"Equally vital are socio-technical and financial capabilities. An engineering graduate must be able to conduct rigorous Techno-Economic Feasibility studies—calculating Net Present Value and payback periods to ensure that an AI investment solves a genuine bottleneck rather than creating expensive digital shelfware. Combined with change management and ethical AI governance, this empowers industrial engineers to lead corporate strategy at the executive table."

Slide 12: Conclusion & Q&A Synthesis (14:15 – 15:00)

- **Slide Visual:** Concluding slide summarizing core takeaways, primary peer-reviewed citations, and team contact information.
- **Speaker 5:**

"In conclusion, Generative AI will not replace Industrial Engineers; rather, Industrial Engineers who master Generative AI will replace those who do not. By offloading computational friction to autonomous agents, we are empowered to design more resilient, sustainable, and human-centric production systems. On behalf of our entire team, thank you for your attention, and we look forward to answering your questions."